BIOL-1 Practice Test 05 — Answer Key

Comprehensive Final Review (Modules 01–15)

Part A: Multiple Choice Answers

Q	Answer
1	C
2	B
3	C
4	B
5	A
6	C
7	C
8	D
9	C
10	B
11	A
12	D
13	D
14	B
15	A
16	A
17	C
18	B
19	A

Q	Answer
20	A
21	B
22	D
23	B
24	B
25	C
26	B
27	A
28	B
29	C
30	C
31	A
32	D
33	A
34	C
35	B
36	C
37	D
38	D

Q	Answer
39	B
40	A
41	B
42	D
43	A
44	B
45	B
46	D
47	C
48	D
49	C
50	B
51	A
52	A
53	C
54	A
55	D
56	D
57	C

Q	Answer
58	C
59	A
60	C
61	D
62	A
63	C
64	D
65	D
66	A
67	D
68	B
69	A
70	B
71	C
72	D
73	A
74	B
75	D

Part B: Fill in the Blank Answers

Q	Acceptable answers (synonyms in parentheses)
76	Control (control group)
77	Homeostasis
78	Electrons (valence electrons / electron pairs — accept “electrons”)
79	Electrons
80	Peptide
81	Lipids (fats / oils / triglycerides — “lipid” alone OK)
82	Lysosomes
83	Nucleus
84	Facilitated (facilitated diffusion; not “active”)
85	Hydrophobic
86	Glycolysis
87	Chlorophyll
88	DNA (DNA polymerase)
89	Codon
90	Anaphase
91	Prophase
92	Phenotype
93	Codominance (incomplete dominance is a different pattern — prefer codominance for “both alleles visible” framing)

Q	Acceptable answers (synonyms in parentheses)
94	Epigenetic
95	Factors (transcription factors)
96	Cloning (recombinant DNA / genetic engineering / molecular cloning)
97	Ligase (DNA ligase)
98	Variation (heritable variation / genetic variation)
99	Convergent
100	Drift (genetic drift)
101	2pq (2 pq)
102	Prezygotic
103	Postzygotic
104	Population
105	Producers (primary producers; autotrophs)

Part C: Short Answer / Free Response Key

Scoring: accept reasonable wording; require the **bold ideas** in each rubric for full credit unless a solid alternative demonstrates the same concept.

106. (Module 01) Two characteristics of life: Any **two** from: **cellular** organization, **reproduction**, **metabolism**, **response** to stimuli, **homeostasis**, **growth/development**, **heritable change** in populations over time / evolution (definitions should match the textbook sense).

107. (Module 01) Theory vs hypothesis: Hypothesis = one testable explanation; **theory** = broad explanatory framework supported by **many** lines of evidence (not “just a guess” in science).

108. (Module 02) Polar water dissolves salt: Water is **polar** partial charges **attract ions** pulled off/away from the crystal (hydration shells / dissociation concept).

109. (Module 02) Buffer: Resists pH change by taking up or releasing $\mathbf{H^+}$ / binds acid or base (bicarbonate buffer in blood is a classic example).

110. (Module 03) Four macromolecules + monomer: Carbohydrates, lipids, proteins, nucleic acids — monomer examples: **monosaccharide**, **glycerol+fatty acids** (or fatty acid), **amino acid**, **nucleotide**.

111. (Module 03) Denatured protein: Loses shape → often loses function; cause: **heat**, **pH** extremes, harsh chemicals, etc.

112. (Module 04) Prokaryote vs eukaryote: Prokaryotes **no membrane nucleus**, smaller/ less compartmentalized; eukaryotes **membrane-bound nucleus** and **organelles** (two clear contrasts).

113. (Module 04) Mitochondria: ATP / energy from **nutrients**; work done mainly in **mitochondrial** compartments (matrix / electron transport / oxidative phosphorylation — intro wording OK).

114. (Module 05) Passive vs active: Passive = with gradient, no ATP for the step (e.g. **diffusion**, **facilitated diffusion**); **Active** = **against** gradient, **uses ATP** (e.g. **sodium–potassium pump**).

115. (Module 05) Very salty outside, water-permeable animal cell: Water moves **out → crenation / shrinkage** (hypertonic outside — framing should match water leaving).

116. (Module 06) Photosynthesis & respiration: Plants fix **carbon/energy** in sugars; respiration **releases** stored energy; **matter** links as $\mathbf{CO_2}$ / **glucose** / $\mathbf{O_2}$ in the big cycle (complementary, not “same process”).

117. (Module 06) Enzyme high temperature: Denatures / unfolds the enzyme → **loses** proper **active site** shape → **slows** or **stops** catalysis.

118. (Module 07) Transcription vs translation: Transcription: **DNA → RNA** (nuclear in euk); **Translation**: **mRNA → protein** at **ribosome** (codon/amino acid link implied).

119. (Module 07) Semiconservative replication: Each new double helix has **one parent** strand and **one new** strand (template-based copying).

120. (Module 08) Mitosis vs meiosis goals: Mitosis: **two** genetically **identical** diploid daughter cells for **growth/repair**; **Meiosis**: **four haploid** gametes with **variation** for **sexual reproduction**.

121. (Module 08) Meiosis variation: Crossing over; **independent assortment**; (also **random fertilization** if not used elsewhere).

122. (Module 09) Aa × Aa phenotype: 3 dominant : 1 recessive (classic **3:1** phenotypes if complete dominance).

123. (Module 09) Heterozygous: Two different alleles; example **Aa**, **Bb**, etc.

124. (Module 10) Epigenetics vs mutation: Mutation = **DNA sequence** change; **epigenetics** = **expression** change **without** changing the **sequence** (often reversible "dials").

125. (Module 10) Environment + trait expression: Example: **nutrition** affecting growth, **stress** / chemicals influencing methylation patterns, **temperature** affecting enzyme activity in **regulation** contexts — must tie to **gene activity without** claiming offspring inherit **changed DNA letters** from gym workouts.

126. (Module 11) GMO + ethics: Foreign gene / engineered organism or crop; ethics: **food safety**, **labeling**, **environment**, **ownership**, **unintended effects** (any two sensible tensions).

127. (Module 11) Gel electrophoresis: Smaller fragments move **farther** / **faster** in the gel than **larger** ones (separation by **size**).

128. (Module 12) Natural selection (friend explanation): Variation is **heritable**; **more offspring** than can survive; **environment** “sorts” who reproduces; **favored** traits become **more common over generations** (four-sentence flow OK).

129. (Module 12) Two evidence lines: e.g. **fossils** (historical change), **homologous** structures (**common ancestry**), **biogeography** (island patterns), **DNA** similarities — each with **one** clause of what it shows.

130. (Module 13) Microevolution + non-selection mechanism: Allele frequency change in a **population**; **genetic drift**, **gene flow**, **mutation** (plus selection — but question asked **other than** selection).

131. (Module 13) Selection modes: Directional favors **one extreme**; **stabilizing** favors **intermediate**; **disruptive** favors **both extremes**.

132. (Module 14) Allopatric vs sympatric: Allopatric: **geographic** barrier then divergence; **Sympatric**: **same** region (e.g. **polyploidy**) new species.

133. (Module 14) Barrier examples: Prezygotic: e.g. **different mating season**, **behavior**, **habitat** split; **Postzygotic**: **hybrid sterile**, **inviable** zygote.

134. (Module 15) Exponential vs logistic: Exponential: rapid **unlimited-phase** growth; **logistic**: growth **slows** as it approaches **carrying capacity K** (resources/limits).

135. (Module 15) 10% rule & chains: ~10% energy **transferred** to **next trophic level** (rest **heat** / metabolism); **more links → more loss → fewer** top predators supported from the same **producer base**.

End of Answer Key